Lesson 25: Single-Subject Research

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 10.1: Single-Subject Research; 10.2: Overview of Single-Subject Research; 10.3: Single-Subject Research Designs.

In this lesson

You will learn how single-subject research studies a few individuals intensively, why its researchers choose it, and how reversal designs, multiple-baseline designs and visual inspection support causal conclusions. In English, you will practise describing change in data over time and sequencing a procedure with *once*, *after* and the past perfect.

Learning goals

After this lesson you should be able to:

- Explain what single-subject research is and how it differs from group research and from case studies.
- Describe reversal and multiple-baseline designs and their variants, and explain how each supports internal validity.
- Judge single-subject data by level, trend and latency, and calculate the percentage of non-overlapping data.
- Describe data over time and sequence a procedure with precise verbs and the past perfect.

Key language

Term or phrase	Meaning	Example
single-subject research	Quantitative research that studies in detail the behaviour of each of a small number of participants (typically 2 to 10; also called small-*n* designs).	Hall and colleagues' six students
group research	Studying many participants and examining their behaviour in terms of means, standard deviations and other group-level statistics.	Most studies in this book
social validity	The extent to which a single-subject study focuses on an intervention with a substantial effect on an important behaviour that can be implemented reliably in real-world contexts.	Teachers found positive attention easy to use.
baseline	A condition in which the dependent variable is measured repeatedly without any treatment.	Robbie studied 25% of the time.

Term or phrase	Meaning	Example
steady state strategy	Letting behaviour become fairly consistent across observations before changing conditions, so a treatment effect is easier to detect.	Wait for a stable baseline.
reversal design (ABA design)	Baseline (A), treatment (B), then baseline again (A); it can be extended (ABAB, ABABA).	Hall's study was ABAB.
multiple-baseline design	Baselines for different participants, dependent variables or settings; the treatment is introduced at a different time on each.	A program started at three schools, 2 weeks apart
visual inspection	Graphing the data and judging whether, and how much, the independent variable affected the dependent variable.	The main analysis method
level; trend	Level: how high or low the dependent variable is in a condition. Trend: gradual increases or decreases across observations.	Studying rose, then levelled off.
latency	The time between the change in conditions and the change in the dependent variable.	A change that begins immediately
percentage of nonoverlapping data (PND)	The percentage of observations in a treatment condition more extreme than the most extreme observation in a relevant baseline condition.	Robbie: 100%
there is little doubt that	A strong conclusion from repeated evidence	"there is little doubt that the treatment was the cause of the improvement"
it is unlikely to be a coincidence	The logic of replication	"...then it is unlikely to be a coincidence"

Rules and explanations

The ideas

The opening study. Hall, Lund and Jackson (1968) recorded every 10 seconds, during a 30-minute period on several days, whether each of six disruptive elementary school students was doing schoolwork (the baseline). Then the teacher gave positive attention ("good work", a pat on the shoulder) whenever the student worked. Robbie went from 25% of his time on schoolwork to 71%. When the attention stopped, all the students studied less; when it was reintroduced, studying rose again (an ABAB design).

What it is, and is not. Single-subject research focuses on objective behaviour through experimental manipulation and control, collects highly structured data and analyses them quantitatively. Case studies and other qualitative research focus on subjective experience, with relatively unstructured data analysed by narrative techniques.

Three assumptions

1. Focus on individuals: group research can hide individual differences (a treatment that helps half the people and harms the other half shows no effect on average), and sometimes one person, such as a particular disruptive student, is the real interest.
2. Discover causal relationships by manipulating an independent variable, measuring a dependent variable carefully and controlling extraneous variables; hence good internal validity.
3. Study strong, consistent effects of biological or social importance (social validity).

Who uses it? Wundt (sensation and consciousness), Ebbinghaus (memory) and Pavlov (classical conditioning) studied a few participants intensively. B. F. Skinner refined the approach in the *experimental analysis of behavior*, mostly with rats and pigeons. Since the 1960s, **applied behavior analysis** has used it with humans on real-world problems (education, developmental disabilities, health). It can serve any theoretical perspective, for example testing a cognitive studying technique, or a clinician documenting a client's improvement.

General features. The dependent variable is measured repeatedly at regular intervals; the study has phases, one condition per phase, labelled A, B, C; conditions change only when behaviour reaches a steady state.

Designs

Design	How it works	Why it supports causal conclusions
Reversal (ABA, ABAB)	Baseline, treatment, removal (and reintroduction)	If behaviour changes with the treatment and changes *back* when it is removed, an extraneous variable is an unlikely cause. (An AB design is only an interrupted time series for one person.)
Multiple-treatment reversal	Baseline (A), positive attention (B), mild punishment (C), baseline, then the treatments in reverse: ABCACB	Compares treatments; the reverse order controls carryover effects
Alternating treatments	Treatments alternate quickly (day by day, or morning and afternoon)	Quick and effective, but only for fast-acting treatments
Multiple baseline across participants, behaviours or settings	Several AB series; the treatment starts at a different time on each	A coincidence would have to happen several times

Multiple-baseline designs avoid two problems of reversal designs: removing a working treatment may be unethical (e.g., one that reduces self-injury in a child), and behaviour may not return to baseline. Ross and Horner (2009) counted the aggressive behaviours of two students who bullied at each of three schools during lunch recess. A bullying prevention program began at the three schools 2 weeks apart, and aggression dropped shortly after it began at each student's school; a holiday or a change in the weather would have to coincide with all three starts. Other versions use one participant's different behaviours (sales calls and reports) or settings (reading at school and at home).

Analysing the data. Visual inspection considers *level*, *trend* (a change in direction is especially telling) and *latency* (a change soon after the conditions change suggests the treatment caused it). Statistics are becoming more common: each participant's mean and standard deviation per condition with *t* tests or analysis of variance, or the PND. All of Robbie's first-treatment measures were above his highest first-baseline measure, a PND of 100%. Statistics are "a supplement to visual inspection, not a replacement for it".

Language focus

1. Describing data over time. Past simple verbs of change, each tied to a phase:

- "The percentage of time he spent studying … was low during the first baseline phase, increased during the first treatment phase until it leveled off, decreased during the second baseline phase, and again increased during the second treatment phase."
- "the number of aggressive behaviors exhibited by each student dropped shortly after the program was implemented"

Useful words: *increase, decrease, drop, rise, level off* (American *leveled*, British *levelled*), *immediately, shortly after, gradually*.

2. Sequencing a procedure.

- Past perfect for the earlier step: "*Once* they *had established* this baseline, they introduced a treatment."
- Time adverbials in order: "*After 2 weeks*, they implemented the program at one school. *After 2 more weeks*, they implemented it at the second school."

> **Note for Arabic speakers** - Arabic uses its past perfect form much less often than English, so learners often leave it out. In a method description, use *had* + past participle to show which past step came first: "Once they had established this baseline, they introduced a treatment."

Practice

1. Reading questions

1. Does "single-subject" mean that only one participant is studied?
2. How does single-subject research differ from a case study?
3. Give two reasons for focusing intensively on individuals.
4. Why did Hall and colleagues' study have good social validity?
5. What is the steady state strategy, and why does it help?
6. Why is an ABA design stronger than an AB design?
7. Which two problems of reversal designs does a multiple-baseline design solve?

2. Applying the section (the book's exercises for this chapter are in Lesson 26)

A. Name the design.

1. Baseline, positive attention, mild punishment for not studying, baseline, punishment, attention.

2. Goal setting is introduced for an office worker's sales calls in Week 5 and for report writing in Week 9.
3. Positive attention for reading is introduced at school first and later at home.
4. Positive attention one day, mild punishment the next, and so on.

B. Visual inspection. *Invented data for practice*: percentage of 10-second intervals spent studying, one session per day.

Participant	Phase A	Phase B	Phase A again
1	20, 25, 22, 18, 24	24, 48, 64, 70, 69	50, 34, 27, 24, 23
2	10, 14, 18, 22, 26	30, 34, 38, 42, 46	(not run)

5. Describe level, trend and latency for Participant 1.
6. Calculate the PND of Participant 1's B phase, compared with the first A phase.
7. Calculate the PND for Participant 2. Does visual inspection agree that the treatment worked? Which principle did the researcher ignore?

3. Language

A. Combine with *Once* + past perfect.

8. The dependent variable reached a steady state. The researcher removed the treatment.
9. The program was implemented at the first school. The researchers waited 2 more weeks before implementing it at the second.

B. 10. Describe Participant 1's data in two or three sentences, using the book's description of Robbie as a model.

Speak with your tutor

- Explain why "the reversal greatly increases the internal validity of the study", and why the self-injury example calls for a multiple-baseline design. Ask for corrections.
- Discuss the ethics of Hall's reversal: was it acceptable to stop the teacher's positive attention to show that it worked?
- PSYC 110 Lecture 4 (Foundations: Skinner) explains that behaviourists studied rats and pigeons because they assumed learning works the same way across species, and that a phrase such as "Good dog" becomes rewarding through conditioning. Discuss how the teacher's "good work" in Hall's study fits this idea.
- Describe Participant 2's data to your tutor without the table and explain why the PND alone was misleading.

Writing task

Your project: a single-subject option. Write 200-260 words describing a single-subject version of your study of sleep and memory. For example, three students each learn a new list of 40 word pairs on

weekday afternoons and recall it after a 90-minute interval: in baseline (A) they rest quietly, and in treatment (B) they have a nap opportunity of up to 90 minutes. The naps begin after 1, 2 and 3 weeks of baseline respectively (a multiple-baseline design across participants). Or use an ABAB design with one student, and say whether recall would really return to baseline when the naps stop. On your own topic, design a reversal or multiple-baseline study of one related behaviour.

Include: the design and why you chose it; the participants; the treatment; an operational definition of the dependent variable, with when and where it is measured; when conditions will change (steady state); how the design protects internal validity; how you will analyse the data (visual inspection of level, trend and latency, and PND).

Self-editing checklist:

- I used the book's name for the design.
- The dependent variable is measured repeatedly, at regular intervals.
- I explained when each phase ends.
- I explained why my results could not easily be a coincidence.
- I used *will* for plans and *once / after* to sequence steps.
- I mentioned level, trend and latency.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PSYC 110 Lecture 4, Foundations: Skinner (video; behaviourism and operant conditioning): https://oyc.yale.edu/psychology/psyc-110/lecture-4

Answer key

1. Reading questions

1. No: it is more typical to study 2 to 10 participants; hence the name small-*n* designs (10.2).
2. A case study is an in-depth, typically qualitative description of an individual; single-subject research studies objective behaviour experimentally, with structured data analysed quantitatively (10.2).
3. Group research can hide individual differences, and sometimes a particular individual's behaviour is the main interest (10.2).
4. It showed strong and consistent effects on a behaviour of obvious importance to teachers, parents and students, and teachers found the treatment easy to implement (10.2).
5. Waiting until behaviour is fairly consistent before changing conditions; any change across conditions is then easier to detect because there is less "noise" (10.3).
6. In an AB design, something else might have changed at the same time. If behaviour also changes back when the treatment is removed, the treatment is much more clearly the cause (10.3).
7. Removing a working treatment may be unethical, and the dependent variable may not return to baseline (10.3).

2. Applying the section

1. Multiple-treatment reversal design (ABCACB).
2. Multiple-baseline design across behaviours.
3. Multiple-baseline design across settings.
4. Alternating treatments design.
5. Level: low in the first baseline (mean 21.8%), much higher in treatment (mean 55%; 64 to 70% in the last three sessions), lower again after removal (mean 31.6%, ending at 24 and 23%). Trend: flat in baseline; rising in treatment until it levels off at about 70%; falling after removal. Latency: short; studying rose from the second treatment session and fell in the first session after removal. The pattern suggests that the treatment was responsible.
6. The highest first-baseline value is 25. Treatment values above 25: 48, 64, 70, 69, so 4 of 5: PND = 80%.
7. All five treatment values exceed the highest baseline value (26), so PND = 100%. Visual inspection disagrees: scores rise by 4 points every session in both phases, so the "improvement" only continues a trend that had begun during baseline. The researcher ignored the steady state strategy: the baseline was still rising when the treatment began. Statistics supplement visual inspection; they do not replace it.

3. Language

8. Once the dependent variable had reached a steady state, the researcher removed the treatment.
9. Once the program had been implemented at the first school, the researchers waited 2 more weeks before implementing it at the second.

10. (example answer) Participant 1's studying was low and stable during the first baseline phase. It increased sharply from the second treatment session until it levelled off at about 70%, and it decreased immediately when the treatment was removed, returning almost to its baseline level.

My notes